IqraSec Academy | WAPT Phase 1

Task 2: Reconnaissance & Enumeration

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Report title	Task 2: Reconnaissance & Enumeration
Rooms / Labs covered	Passive Reconnaissance; Active Reconnaissance; Content Discovery (TryHackMe)
Date completed	1 November 2026

1. Objective

The objective of this task was to learn how to map a target's attack surface before testing it — identifying hosts, open services, and hidden content — using both passive methods (no direct interaction with the target) and active methods (direct probing such as ping, traceroute and netcat).

2. Concept / Understanding

Passive reconnaissance

- Passive recon gathers information about a target without directly interacting with it, for example using **whois** to look up domain registration details, and **nslookup**/**dig** to resolve DNS records.
- Because no packets are sent to the target itself, passive recon is far less likely to be logged or to alert the target.

Active reconnaissance

- Active recon interacts directly with the target: **ping** (ICMP) checks whether a host is reachable and measures round-trip time; **traceroute** maps the network hops between the source and the destination; **telnet**/**netcat** can connect directly to a TCP port and read the service banner it returns.
- Different traceroute modes exist: UDP (default on Linux), TCP (traceroute -T, useful for bypassing UDP filtering) and ICMP (traceroute -I). A route showing * * * for every hop usually means the intermediate routers or a firewall are dropping or not responding to that probe type.
- Netcat (nc) can be used as a simple TCP client to connect to a service port (e.g. FTP on port 21) and read back its banner, which often reveals the exact server software and version in use.

Content discovery

- Hidden web content — files, directories, backup copies, admin panels — can be found manually (checking common paths), through OSINT, or with automated tools such as Gobuster that brute-force paths against a wordlist.
- Response headers (e.g. Server, X-Powered-By) and framework-specific files can reveal the technology stack running on a target.

3. Practical Work

- Completed the **Passive Reconnaissance** room (100%), covering whois, nslookup and dig.

- Completed the **Content Discovery** room (100%), covering manual discovery of common files and headers/framework identification.
- Worked through the **Active Reconnaissance** room (100%, Tasks 1–7: Introduction, Web Browser, Ping, Traceroute, Telnet, Netcat, Putting It All Together) on the TryHackMe AttackBox against the room's target machines (10.112.132.132 and 10.112.165.183):
- Used ping and ping -n against the target, confirming 10/10 packets received with 0% loss, and worked out the ICMP header details (echo request size flag -s, 8-byte ICMP header).
- Ran traceroute against the target IP directly (reached in a single hop, since it sits on the same internal subnet as the AttackBox) and against tryhackme.com over the public internet (30 hops, every hop showing * * * — the ICMP probes were being filtered before reaching tryhackme.com).
- Compared the room's reference Traceroute A and Traceroute B outputs to answer the "last hop before reaching tryhackme.com" questions (100.92.9.83 and 99.83.89.19).
- Used Netcat to connect to the target VM's FTP service on port 21 and read the banner to identify the running server version (vsftpd 0.17).

4. Tools Used

Tool	Purpose
TryHackMe	Completed the guided rooms and AttackBox exercises
ping / ping -n	Checked host reachability and inspected ICMP packets
traceroute (UDP/TCP/ICMP modes)	Mapped network hops to the target and to tryhackme.com
netcat (nc)	Connected to the FTP port and read the service banner
whois, nslookup, dig	Passive recon – domain and DNS information gathering

5. Evidence

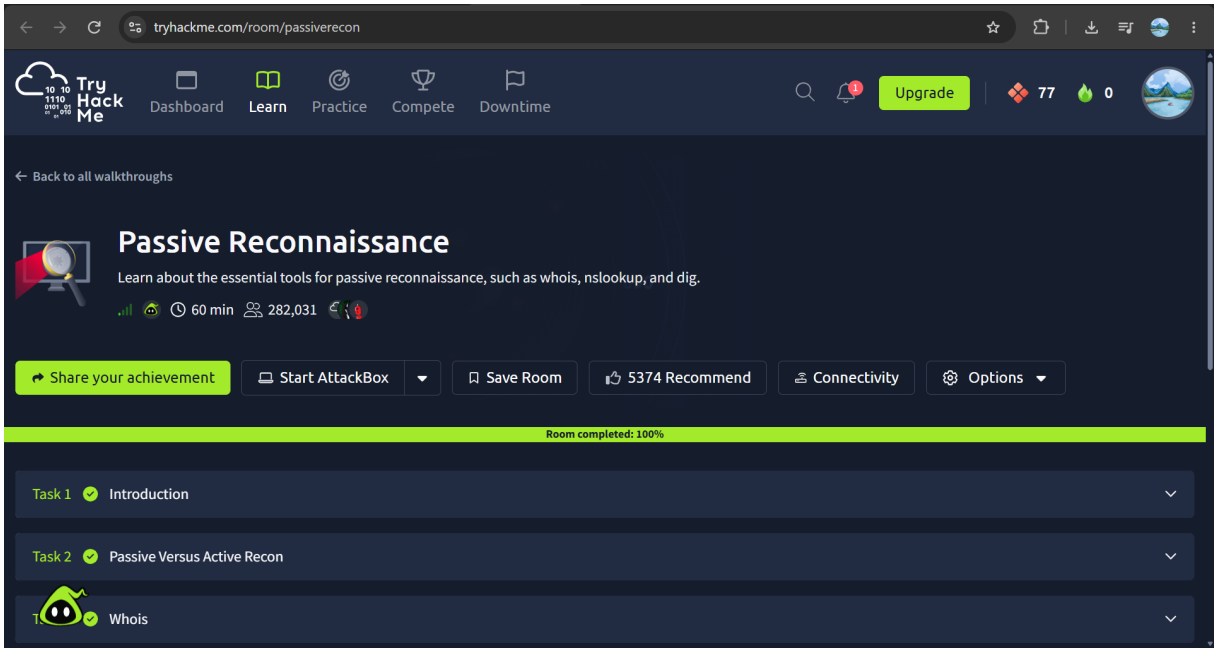


Figure 1: *Passive Reconnaissance room completed at 100%.*

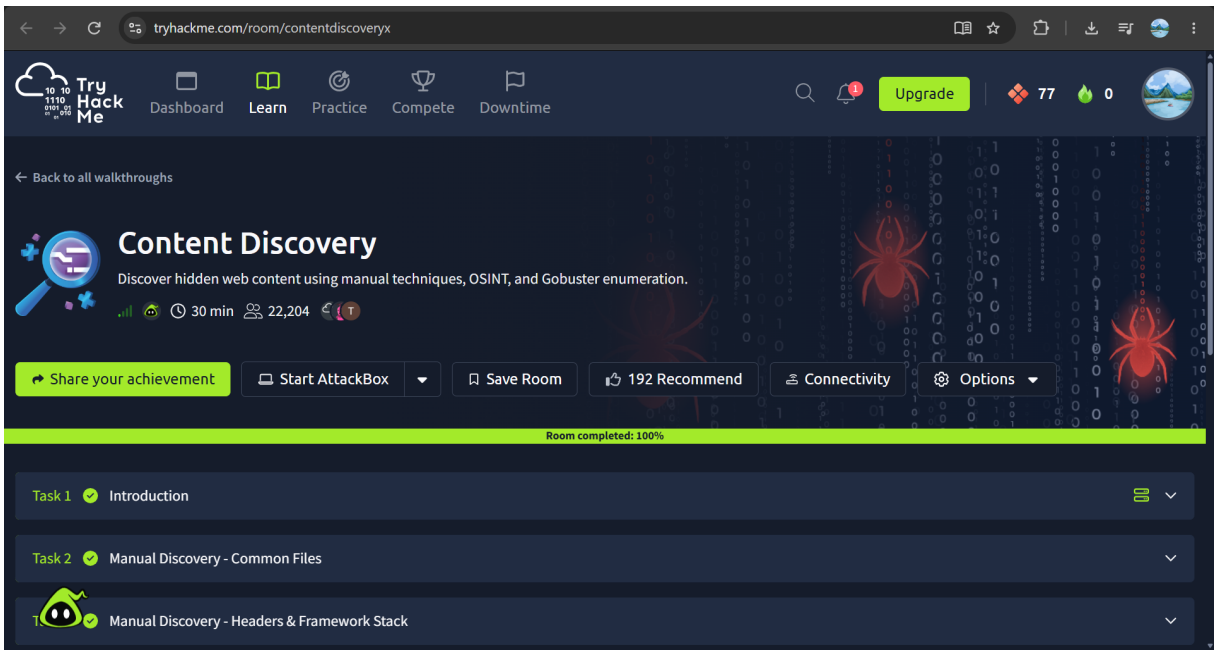


Figure 2: *Content Discovery room completed at 100%.*

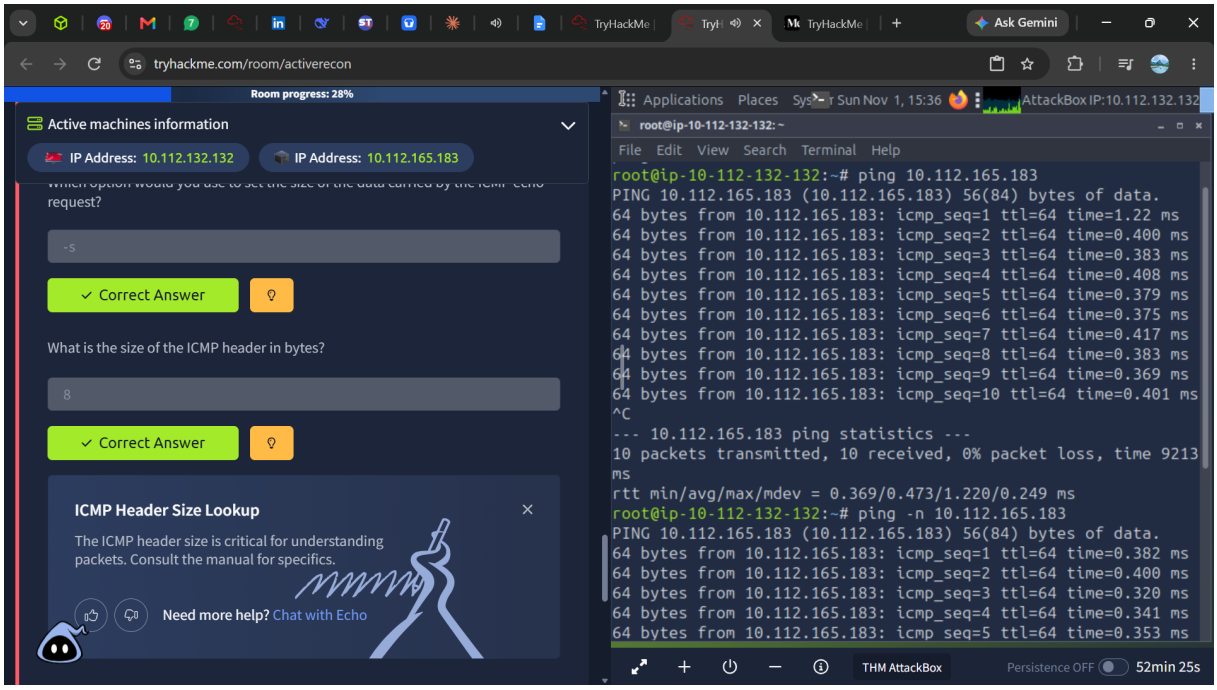


Figure 3: *ping and ping -n against 10.112.165.183 — 10/10 packets received, 0% loss; ICMP header questions answered.*

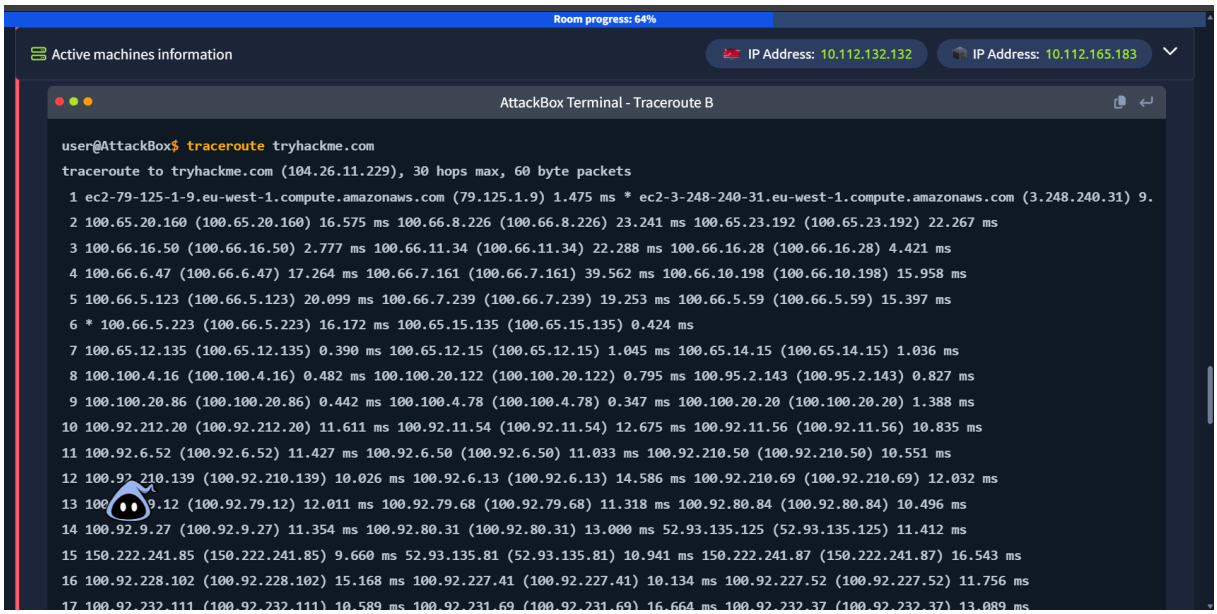


Figure 4: *traceroute tryhackme.com (Traceroute B reference output) used to work out the last router before the destination.*

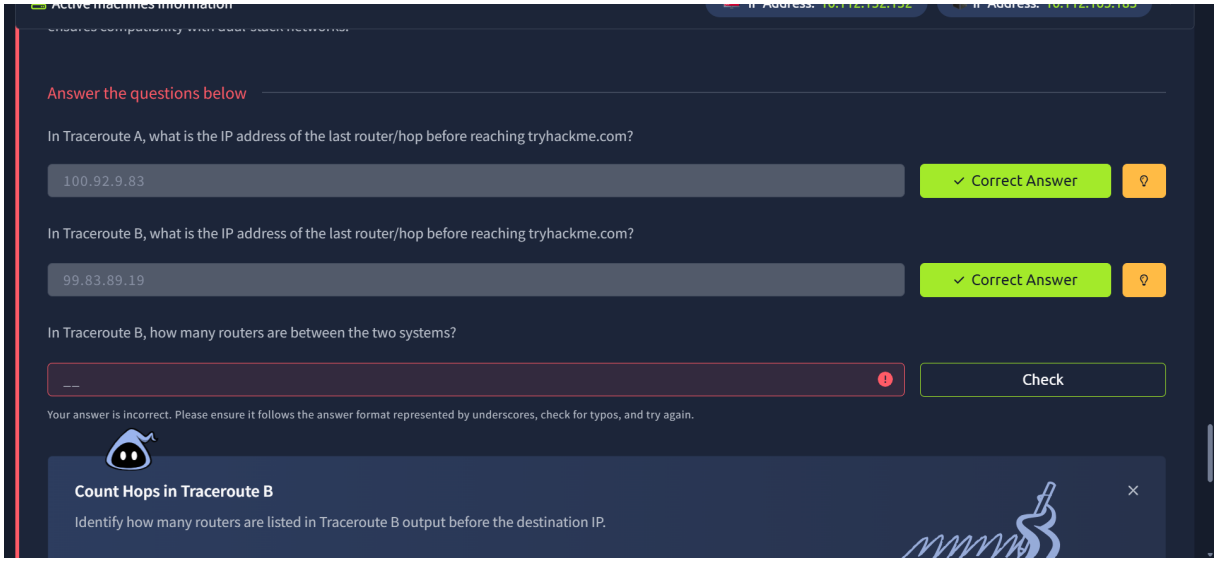


Figure 5: *Correct answers submitted for Traceroute A (100.92.9.83) and Traceroute B (99.83.89.19) last-hop questions.*

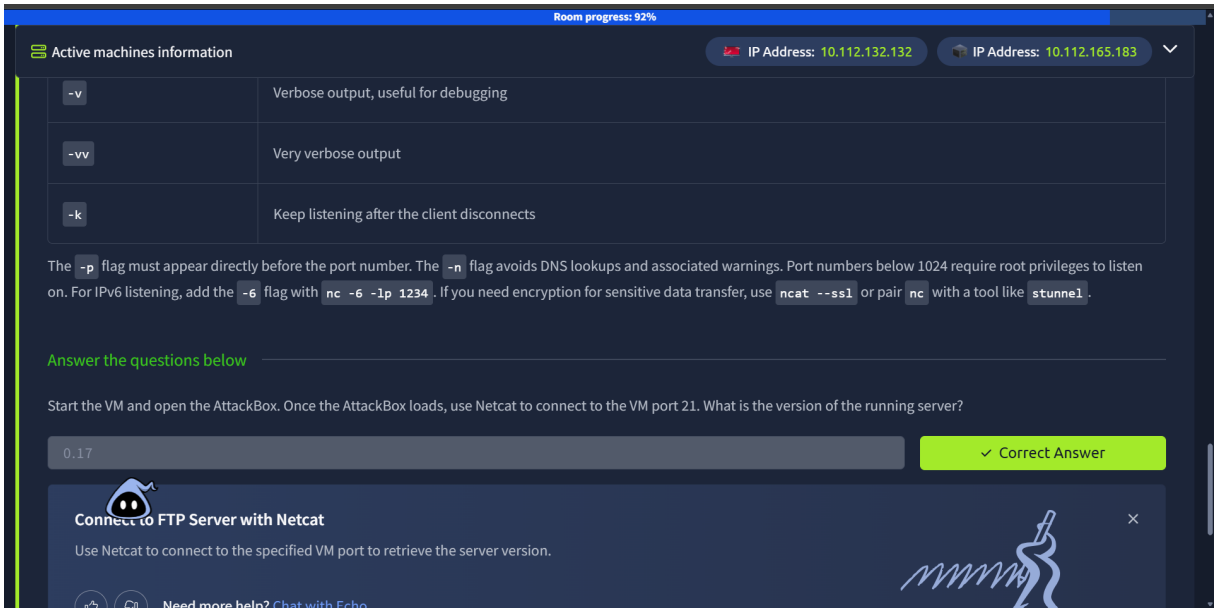


Figure 6: *Netcat used to connect to the target's FTP port (21); server version identified as 0.17 (vsftpd).*

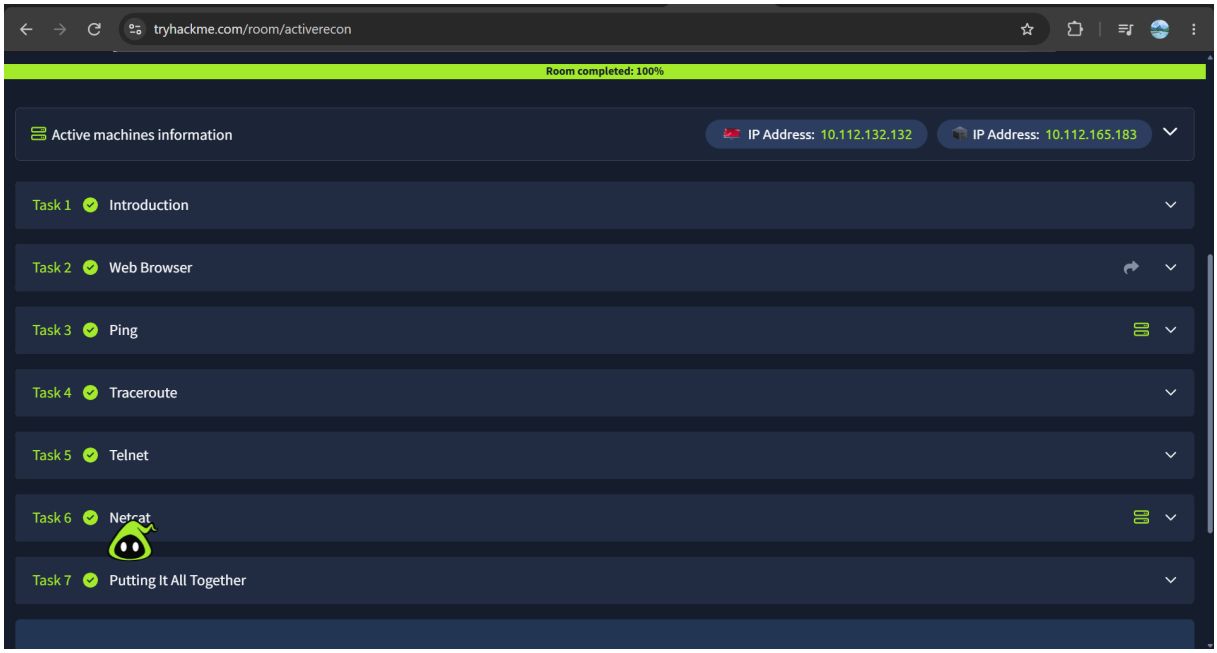


Figure 7: *Active Reconnaissance room completed at 100% (Tasks 1-7 all ticked).*

■ Evidence gap: No screenshot shows the actual whois/nslookup/dig commands and output from the Passive Reconnaissance room, or a Gobuster/manual-discovery result from Content Discovery — only their completion pages are included. Add a command + output screenshot from each if possible, since the guide asks for commands/techniques used, not just completion screens.

6. Result

All three rooms were completed. The target's reachability and ICMP behaviour were confirmed with ping, the network path to the target and to tryhackme.com was mapped with traceroute in three different modes, and a running service version was fingerprinted directly with netcat. This demonstrates the difference between passive information gathering and active probing, and how both feed into building a picture of a target's attack surface.

7. Key Learnings

- Passive recon (whois, DNS lookups) gathers useful information without touching the target directly, while active recon (ping, traceroute, netcat) interacts with it and can be logged.
- A traceroute returning * * * at every hop does not always mean the host is down — it often means ICMP or UDP probes are being filtered somewhere on the path.
- Service banners grabbed with netcat can reveal exact software versions, which is valuable for identifying known vulnerabilities later.
- Content discovery (manual checks, headers, automated brute-forcing) is how hidden endpoints and the underlying technology stack are found before deeper testing begins.

Declaration: I confirm that this report documents my own work and that all screenshots were taken by me during the tasks described.

Name: Maryam Batool Date: 1 November 2026